Frank Nielsen, Sony CSL. Selected publications (2023-2026)

Journals

- Geometric Jensen-Shannon divergence between Gaussian measures on Hilbert space, IEEE Transactions on Information Theory (accepted), 2026
- Equivariant log-extrinsic statistics on symmetric cones, Information Geometry 2026
- Generalized Legendre Transforms Have Roots in Information Geometry, Entropy 2026
- Mirror Descent and Exponentiated Gradient Algorithms Using Trace-Form Entropies, Entropy 2025
- A geometric modeling of Occam's razor in deep learning, Information Geometry 2025
- Two types of geometric Jensen–Shannon divergences, Entropy 2025
- Fast Proxy Centers for the Jeffreys Centroid: The Jeffreys–Fisher–Rao Center and the Gauss–Bregman Inductive Center, Entropy 2024
- Information measures and geometry of the hyperbolic exponential families of Poincaré and hyperboloid distributions, Information Geometry 2024
- Variational representations of annealing paths: Bregman information under monotonic embedding, Information Geometry 2024
- On the f-divergences between densities of a multivariate location or scale family, Statistics and Computing 2024
- Symplectic Bregman divergences, Entropy 2024
- Divergences Induced by the Cumulant and Partition Functions of Exponential Families and Their Deformations Induced by Comparative Convexity, Entropy 2024
- What is... an inductive mean?, Notices of the American Mathematical Society 2023
- A simple approximation method for the Fisher–Rao distance between multivariate normal distributions, Entropy 2023
- On f-Divergences Between Cauchy Distributions, IEEE Transactions Information Theory 2023

Conferences

- Curved representational Bregman divergences and their applications, GSI 2025
- A Rate-Distortion View of Uncertainty Quantification, ICML 2024
- Hyperbolic Embeddings of Supervised Models, NeurIPS 2024
- Optimal transport with tempered exponential measures, AAAI 2024
- Simplifying momentum-based positive-definite submanifold optimization with applications to deep learning, ICML 2023
- Quasi-arithmetic Centers, Quasi-arithmetic Mixtures, and the Jensen-Shannon-

Divergences, GSI 2023

- Partial K-Means with M Outliers: Mathematical Programs and Complexity Results, International Conference on Optimization and Learning 2023

Book chapter

Approximation and bounding techniques for the Fisher-Rao distances, Handbook of Statistics 2024

Workshops

- Hilbert geometry of the symmetric positive-definite bicone, NeurIPS Workshop on Symmetry and Geometry in Neural Representations, NeurReps 2025
- A New Geometric Approach of Adaptive Neighborhood Selection for Classification, NeurIPS Workshop on Symmetry and Geometry in Neural Representations , NeurReps 2024
- Fisher-Rao and pullback Hilbert cone distances on the multivariate Gaussian manifold with applications to simplification and quantization of mixtures, 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG), 2023
- Non-linear Embeddings in Hilbert Simplex Geometry, 2nd Annual Workshop on Topology, Algebra, and Geometry in Machine Learning (TAG), 2023

Technical reports

- Dual-connection midpoint matrix means and their Gauss composition, arXiv:2609.07551
- Minimum enclosing Bregman balls made easy, arXiv:2607.24197
- Group invariance of f-divergences and the Fisher-Rao distance, arXiv:2606.25790
- Neural Legendre-Fenchel transform with Hessian Preconditioning, arXiv:2606.09077
- Generalising maximum mean discrepancy: kernelised functional Bregman divergences, arXiv:2604.24047
- Quadratic polarity and polar Fenchel-Young divergences from the canonical Legendre polarity, arXiv:2603.04812
- Geometric structures and deviations on James' symmetric positive-definite matrix bicone domain, arXiv:2603.02483
- Scales of Fréchet means and Karcher quasi-arithmetic means, arXiv:2511.21173
- Hilbert geometry of the symmetric positive-definite bicone: Application to the geometry of the extended Gaussian family, arXiv:2508.14369
- pyBregMan: A Python library for Bregman Manifolds, arXiv:2408.04175
- Tempered Calculus for ML: Application to Hyperbolic Model Embedding, arXiv:2402.04163